Supernature: the plan

Build Queue item 96 · Claudia · October 13, 2026 · **Status: planned, not built.** Building starts when her assets arrive and she answers the short list in section 12.

Supernature is a new app on the student computer. It replaces the old idea of a "Potion Lab". Students discover real mythical creatures from around the world, make their own creatures, and build a small living world where those creatures have to eat, rest and get along. Potions and wand spells are the magic that ties it together. Everything is 2D pixel art. Nothing is shared between students. **There is no blood anywhere: not in Supernature and not in any other part of the platform.**

Her words that set the direction (2026-10-13): "creating student own supernatural creatures and their interaction with the ecosystem the student creates seems to be the priority. these are ideas from students, claudia should explore connections, patterns, and actual game play logic." Then: "claudia, make sure this is a cohesive, dev plan before going. ill get you assets."

1. The one big idea

Supernature works like a field naturalist's journal that happens to be magic. Every part feeds the next part, so a student always has a reason to go somewhere else in the app:

Room	What the student does	What it gives the other rooms
Globe	Spins the world, taps a continent, ocean or sea, zooms in to countries, finds glowing creature pins	Unlocks Bestiary pages, earns ingredients ("Earned coral by exploring the Mermaid")
Bestiary (the creature book)	Reads each creature's page; unknown creatures are "???" silhouettes; sorts trait cards to finish a page	Real creatures to remix; trait words to build with
Creature Maker	Explores, remixes, or builds a creature from scratch out of pixel parts and trait cards	Creatures to put in a Habitat
Habitat (the priority)	Picks a biome, adds plants, real animals and creatures, runs the seasons, keeps the food web in balance	Adaptation traits, creature levels, Naturalist rank
Cauldron	Reads a recipe and follows it in order, mixes, then traces the spell word in cursive with a wand	Potions for pets, for their own character in Town Square, and for Habitat problems
Supernature Shop	Buys ingredients with their real dashboard money	Ingredients for the Cauldron

The loop a student lives in: **find it, learn it, make one, keep it alive, brew for it.**

2. Learning goals and where each one lives

- **Mythology and culture** (Globe, Bestiary): real creatures, the country and people whose stories they come from, told respectfully (section 9).

- **Ecosystems are fragile** (Habitat): producers, herbivores, carnivores, omnivores and decomposers; food chains and the food web drawn as arrows; what happens when one part is removed.
- **Geography** (Globe): continents, the five oceans and big seas, then countries.
- **Vocabulary through sorting traits** (Bestiary, Creature Maker, Habitat): her list, grouped so students sort them:
  - Body: sharp, sword-like hands, one eye, legs, flippers, wings.
  - Moving: walking, flying, swimming (aquatic), burrowing.
  - Survival: water breathing, regenerative, immunity to pain, endurance (the cockroach is the champion).
  - Mind: an intelligence scale (from "follows its senses" up to "makes fire and tools"), mischievous, curious, gentle.
  - Social: pack hunter or loner, cooperation.
  - Adaptations: traits learned from where the creature lives (section 5).
- **Reading is a job** (Cauldron, Bestiary): every recipe and creature page is read; read-aloud on every screen.
- **Order of operations** (Cauldron): recipe steps happen in order, and anything in the small mixing bowl happens first, just like parentheses in math (section 6).
- **Cursive** (Wand): tracing spell words in Cursive Without Tears letter shapes (section 7).
- **Self-regulation** (everywhere): the Energy scale and Zones (section 8).

Questions from question sets are **not** part of Supernature for now (her answer: "no questions yet").

3. Globe

- A pixel-art globe students spin with a swipe and zoom with a pinch. It turns slowly on its own when nobody is touching it.
- **Zoomed out:** the seven continents, the five oceans (Pacific, Atlantic, Indian, Southern, Arctic) and the big seas (Mediterranean, Caribbean, South China Sea, North Sea and a few more). The one under a finger lifts and glows (touch, not hover, since students are on iPads). Its name is said aloud.
- **Zoomed in:** a flat pixel map of that continent or ocean with country borders. A country with a discovered or discoverable creature has a pin. Pins for creatures not found yet show "?".
- Tapping a pin opens the Bestiary page for that creature. The first visit earns that creature's ingredient.
- **Antarctica** has no traditional creature stories, because no people lived there to tell them. That fact is taught on the page, and it is the open frontier: students may place their own creatures there.
- Built from public domain world map data (Natural Earth), drawn down to a pixel grid, so borders are real.

4. Bestiary (the creature book)

- A pixel spell book with page turns, like her reference picture. Pages are grouped by continent and ocean, then a **My Creatures** section at the back.
- An undiscovered creature is a dark silhouette with "???" and only its region shown.
- A discovered page shows: the pixel picture, its name and how to say it (read aloud), where it comes from (country and people), a short story told respectfully, its traits, its diet, its habitat, and an Energy face.
- **Finish the page job (vocabulary):** trait cards are mixed up beside the page; the student drags each to "has it" or "does not have it", using the story they just read. A finished page gets a gold seal. Kind help on a wrong sort, never a buzzer.
- Story pages read themselves aloud by default with a 🔊/🔇 toggle (her rule for story screens).

5. Creature Maker and Habitat (the priority)

Creature Maker

Four ways in, her list:

1. **Explore a real creature:** read it and see it move. Read only.
2. **Remix a real creature:** start from a real one and swap parts and traits. The card says "Inspired by the Griffin (Greece)". Only creatures marked "remix ok" can be remixed (section 9).
3. **Build from scratch:** pick a body plan (walker, flier, swimmer, slitherer, floater), then pixel parts: head, eyes (1, 2 or 3), arms (hands, claws, sword-hands, flippers, tentacles), covering (fur, scales, feathers, shell, slime), tail, wings, colors.
4. **Spark:** a random start for students who freeze at a blank page.

Then the student:

- Sees **body traits filled in automatically** from the parts (three eyes gives "sharp sight"; flippers give "aquatic"). These are the traits "assigned to the creature".
- **Chooses personality** (mischievous, gentle, curious, bossy, shy), **social style** (pack or loner), **intelligence** (1 to 5) and **diet** (plants, meat, both, leftovers, or a magic food like moonlight or dreams).
- **Names it** and writes one sentence about it. The writing check is gentle: a capital letter at the start, punctuation at the end (the same habit as the gym form).
- The creature goes into My Creatures in the Bestiary.

Habitat: the actual game play logic

This is where students' creatures interact with a world they made.

1. **Pick a biome:** forest, ocean, desert, mountain, swamp, tundra, grassland, or sky islands. Each biome has sun, water and cold levels.

2. **Place life:** producers (grass, trees, kelp, plankton, glowing mushrooms), real animals (rabbits, deer, fish, beetles, cockroaches), and up to a few of their creatures. Each is a pixel sprite that walks, swims or flies around.
3. **Run a season:** the student taps "Next season". Each season:
    - Producers grow, depending on the biome's sun and water.
    - Herbivores eat producers; carnivores eat herbivores; omnivores eat either; decomposers turn leftovers back into soil, which helps producers.
    - Each kind's group gets bigger, stays the same or gets smaller, shown as pixel counters (1 to 10 icons).
    - **Eating is never shown.** A predator "catches" and a little poof of smoke shows the prey running back to its burrow; the number goes down. A group that runs out of food "packs up and moves away" (a tiny suitcase walking off screen). No blood, no hurt animals, ever.
4. **The food web** is a tap away: arrows from each food to whatever eats it (arrows show where the energy goes). Students can turn on "what if": hide one kind and watch a ripple show who is affected. This is the fragility lesson.
5. **Balance meter:** Balanced, Wobbly or Falling apart, with one plain sentence saying why ("Too many hunters and not enough rabbits.").
6. **Traits matter:** a creature without water breathing struggles in the ocean; pack hunters catch bigger food but need more of it; regenerative creatures bounce back after a hard season; endurance (the cockroach trait) lets a kind survive a bad season; a fire-making creature can warm the tundra. Students learn which traits fit which places by trying.
7. **Adaptations are learned:** a creature that lives through three seasons in a biome may learn one of that biome's adaptations (thick fur in the tundra, night vision in a cave, camouflage in the forest). The student picks the one that fits; it is added to the creature's page as "learned".
8. **Levels:** creatures gain a level after each balanced season. The student earns a **Naturalist rank** (Apprentice, Field Scout, Creature Keeper, Habitat Builder, Grand Naturalist) that unlocks new biomes, parts and recipes.

Saved per student and private: their creatures, habitats, discovered pages, ingredients and potions.

6. Cauldron (potions)

- **The Cookbook:** about 10 base recipes to start (her number). Ideas: Flying, Strength, Speed, Glow, Tiny, Giant, Calm, Bounce, Invisible (a see-through shimmer), Grow (for plants in a Habitat).
- **Discovered recipes:** brewing a base recipe unlocks a harder version with bigger needs: Flying unlocks Super Speed Flying; Glow unlocks Rainbow Glow; Strength unlocks Mountain Strength.
- **Reading is the job:** the student reads the recipe card (read aloud on tap) and follows it.
- **Order of operations:** each recipe is a list of steps: add, stir (a number of times), heat, wait, wand. Anything in the **small mixing bowl** happens first, like parentheses. Example: "Mix (grass + dew) in the bowl. Pour it in. Add the feather. Stir 3 times. Trace *fly*." Doing a step out of order is never a fail: the

cauldron fizzles with a silly puff of smoke and a burp, the step list shows which step comes first, and they try again.

- **Big magic effects:** smoke poofs, sparkles, bubbling, color swirls, a lid that rattles. Calm mode and reduced motion turn these down.
- **Where potions go:**
  - **Pets:** a Strength potion makes a pet glow and lift a tiny dumbbell. The effect is for fun and lasts a short time; it does not replace care or gym training.
  - **The student's own character:** a Flying potion lets their Town Square character float and fly for a minute; Tiny, Giant, Glow and Bounce work the same way.
  - **Habitats:** Grow helps a struggling plant group; Calm brings a creature's energy back to just right.

7. Wands and cursive

- The last step of every recipe is the spell: the student traces the spell word with a wand (a finger or an Apple Pencil). Sparkles follow the stroke.
- Letters follow **Cursive Without Tears** shapes and stroke order: a start dot, a guide path, and a gentle check that the stroke went the right way. Close enough counts.
- Spell words only use letters the class has already learned (see section 12).
- Spell words grow with rank: *fly* and *glow* first, then *bounce* and *giant*, then *super speed*.
- Spell tracing can also be practiced on its own from the Cauldron's spell book.

8. Self-regulation: Energy and Zones

From her MyEnergy page (Autism Level UP!), which uses an energy scale from very high to sleepy:

- Every creature has an **Energy face** on a 5-step scale: very high (frazzled), high (excited), just right, low (droopy), very low (sleepy). We draw our own pixel faces in the same spirit; their artwork is not copied.
- In the Habitat, energy is a game state students learn to read: crowded or overfed creatures run too high, hungry ones run low. The goal is "just right", and students fix it with space, food, rest or a Calm potion. This models the idea without ever telling a student how they feel.
- When a student opens Supernature, they can do an optional quick **My Energy check-in** on the same scale. The result can suggest a matching potion ("Feeling too high? Try brewing a Calm potion."), and calm potions use their real calming tools as recipe steps (the stir is 5 slow breaths with a breathing circle).
- Creature pages show a Zone too, using the Zones colors the dashboard already uses, so the two systems stay connected and never compete.
- **Level up:** creatures level up, students earn Naturalist ranks, and new parts, biomes and recipes unlock in small, clear steps with a "what unlocks next" bar.

9. Culture and respect: the rules and locks

Her rule: "prioritize culturally responsive teaching practices and add parameters and locks around culturally appropriative game play."

- **She approves every region's list** in a teacher screen before any creature appears to students. Each creature has three settings: **Hidden**, **Explore only**, or **Remix ok**.
- **Explore only** is the default for figures that are sacred in a living religion or belong to Indigenous peoples. Students read and learn about them; they cannot remix them, brew from them or put them in a Habitat.
- **Remix ok** is for creatures from old stories that are now shared widely (the Griffin, the Kraken, the Jackalope) and for gods and goddesses of ancient religions no longer practiced as a living faith (Greek, Norse, Roman, ancient Egyptian). She asked to keep gods and goddesses; this is how they stay in respectfully.
- **Left out entirely:** creatures a community has asked outsiders not to tell (the Skinwalker and the Wendigo), and creatures whose story is about blood (the Chupacabra and vampires).
- Every page names the country and the people the story comes from, uses the creature's own name first, and says "In stories told by...", not "This creature is real" or "This creature is fake".
- Ingredients earned from a creature are respectful (coral from the Mermaid, a lost feather from the Griffin); nothing is taken from sacred figures.
- **No sharing:** no gallery, no sending, no showing other students' creatures. She can see every student's book from the teacher side.

Starter creature list for her approval (proposed setting in brackets)

- **Europe:** Griffin, Greece [remix]; Pegasus, Greece [remix]; Cyclops, Greece [remix]; Hydra, Greece, regenerative heads, no blood [remix]; Unicorn, Scotland [remix]; Loch Ness Monster, Scotland [remix]; Selkie, Scotland and Ireland [remix]; Zeus, Athena, Poseidon, Thor, Odin, Freya [remix].
- **Asia:** Long, the Chinese dragon, China [explore]; Fenghuang, China [explore]; Qilin, China [explore]; Kappa, Japan, mischievous and aquatic [remix]; Tanuki, Japan [remix]; Baku, Japan, eats bad dreams [remix]; Haetae, Korea [remix]; Simurgh, Iran [explore]; Naga and Garuda, India and Southeast Asia [explore].
- **Africa:** Anansi, Ghana (Akan people), the trickster spider [explore]; Sphinx and Bennu, Egypt [remix]; Ra, Bastet, Anubis, ancient Egypt [remix]; Mokele-mbembe, Congo [remix]; Grootslang, South Africa [remix].
- **North America:** Thunderbird, many Indigenous nations [explore]; Sasquatch, a Salish name [explore]; Jackalope, Wyoming, USA [remix]; Mothman, West Virginia, USA [remix]; Ogopogo, Canada (n'ha-a-itk of the Syilx people) [explore].
- **South America:** Curupira, Brazil, forest guardian with backward feet [explore]; Iara, Brazil, river mermaid [explore]; Boto, the Amazon river dolphin [explore]; Alicanto, Chile, a bird that eats gold and silver [remix]; Quetzalcoatl, Mexico (Aztec) [explore].

- **Oceania:** Bunyip, Australia (Aboriginal peoples) [explore]; Rainbow Serpent, Australia (Aboriginal peoples) [explore]; Taniwha, New Zealand (Maori) [explore]; Drop Bear, Australia, a joke creature [remix]; Yowie, Australia [remix].
- **Oceans and seas:** Kraken, North Atlantic [remix]; Mermaids, worldwide [remix]; Charybdis, Mediterranean [remix]; Umibozu, Sea of Japan [remix]; sea serpents [remix].
- **Antarctica:** no traditional creatures; the students' frontier.

10. Ingredients and the Supernature Shop

- **Free basics:** glass bottle, grass, water, dew, pebble, salt, a feather, moonlight, a leaf, sand. Never gross.
- **Shop ingredients ($1 to $10):** eye of newt (fun fact on the label: in old recipes this was really a mustard seed), hair of the dog, toad burp, pixie dust, troll toenail, dragon sneeze, frog giggle, bat whisper. Gross humor is allowed here but kept to a few (her rule: limited, never in the basics).
- **Super rare ($15 to $50):** phoenix ember, kraken ink, unicorn sparkle, star sap, thunder in a jar.
- Paid with their real dashboard money, the same money as the Marketplace, with a receipt line in their bank history. Prices show with commas.
- **Earned by exploring:** each real creature's page gives one ingredient the first time ("Earned coral by exploring the Mermaid").

11. Other changes that come with it

- **Science Lab:** the fantasy or magic mode is removed; it becomes chemistry only (her rule: "drop fantasy mode").
- **Alchemy:** stays exactly the same.
- **Computer app:** a new "Supernature" icon on the student computer. The host character is to be decided; she is sending the art.
- **Every screen** has read-aloud; story pages read themselves aloud by default with a toggle; voices go through the human voice service.
- **iPad first:** swipe and pinch on the globe, big drag targets, no hover-only actions, both orientations, nothing overlapping.
- **Bug and idea Report** bubble in the corner, like every game.

12. What we need from her before building

Assets (she said she will send them):

1. The host character (any pose; a few expressions if possible).
2. Pixel art for real creatures, if she has a pack. If not, Claudia builds them from the same pixel parts the Creature Maker uses.

3. Optional: a pixel book and page set, cauldron, wand, potion bottles, biome tiles, smoke and sparkle effects. Anything missing gets drawn in the same pixel style.

Answers (only four):

1. **The starter list:** approve, change, or remove creatures in section 9, including the Explore only and Remix ok marks.
2. **Cursive:** which letters has the class learned so far in Cursive Without Tears? Spell words will only use those, and grow as the class learns more.
3. **Energy check-in:** OK to use a 5-step energy scale with our own pixel faces, and to offer the optional My Energy check-in when Supernature opens?
4. **Potion effects on pets:** for fun only (a glow and a short animation), or should a Strength potion also count toward gym training?

13. Build order (each step ships on its own)

1. **Foundation:** Science Lab loses fantasy mode; Supernature app on the computer; saved data; the teacher approval screen for creatures (everything Hidden until she approves).
2. **Creature Maker and Habitat** (her priority), with a simple list of approved real creatures to explore and remix.
3. **Bestiary:** the pixel book, "???" pages, the trait sorting job, My Creatures.
4. **Globe:** spin, zoom, continents, oceans and seas, countries, pins, earned ingredients.
5. **Cauldron and Shop:** the Cookbook, order of operations, discovered recipes, effects, the ingredient economy.
6. **Wand cursive** spells.
7. **Potions out in the world:** pets, the student's Town Square character, Habitats.
8. **Energy and level up** threaded through all of it (started in step 2, finished here).